COMPUTER SCIENCE
YEARS 1-9
CURRICULUM FRAMEWORK

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Nature of the Subject

Computer Science is a dynamic subject that aims to develop students' systematic problem-solving skills while fostering an understanding of digital technology and its role in a technology-driven society. The subject helps students cultivate computational thinking, encouraging creativity and innovation as they tackle complex challenges. Through exposure to computer architecture, coding, and digital media, students not only gain technical proficiency but also enhance their ability to express ideas and collaborate effectively. Additionally, Computer Science emphasises wellbeing, preparing students to navigate and contribute positively to a rapidly evolving technological landscape with resilience and adaptability.

Conceptual Framework

St George's Core

Concept	Outline
Form:	Refers to the structure, organisation, and representation of data, code, or systems, independent of specific content. It encompasses how information is arranged, displayed, or processed, influencing usability, efficiency, and interaction. Form can apply to user interfaces, data models, programming paradigms, and system architectures, shaping the way users and systems engage with information.
Adaptation:	The capacity of computational systems and solutions to adjust their logic, structure, or behaviour in response to evolving data, conditions, or requirements. In Computer Science, adaptation is often achieved through iterative processes, where repeated cycles of development, testing, and refinement enable systems to respond dynamically and remain effective in changing contexts.
Function:	The purpose behind designing digital systems or writing code, ensuring solutions meet user needs and solve specific challenges effectively.

Subject Specific

Concept	Outline
Logic	The principles of reasoning that guide algorithms and programming, forming the foundation for creating reliable, efficient digital solutions.
Procedure	A sequence of instructions that performs a task without returning a value, essential for structuring code and achieving specific outcomes in programming.
Process	A running program executing tasks by following instructions, using system resources and memory, and representing how programs interact with a computer's hardware.
Data	Raw information in various forms (numbers, text, images) that can be collected, stored, and processed to generate meaningful outputs.
Innovation	The development and application of new ideas, products, or techniques in technology, driving progress and solving challenges in Computer Science.

Key Skills in Computer Science

Skill	Outline	Skills for Learning
Programming	Developing the ability to design, write, test, and refine code to solve tasks or create systems, integrating planning and structured design to ensure efficiency and clarity.	
Data Handling	Learning to collect, organise, analyse, and interpret data, presenting findings in meaningful ways to inform decision-making and problem-solving.	Communication
Digital Literacy	Building skills to navigate, evaluate, and create digital content responsibly, with a focus on understanding and using emerging technologies like artificial intelligence. It fosters effective communication, ethical digital practices, critical thinking, and a strong foundation in online safety, preparing individuals to engage responsibly in an increasingly digital world.	Communication Social Research
Computational Thinking	A problem-solving skill that involves breaking down complex problems into smaller, manageable parts, applying logical reasoning, and recognising patterns. It enables learners to design algorithms, abstract key elements, and iteratively refine solutions, fostering creativity and efficiency in developing computational solutions.	Thinking
Creativity	The ability to envision and develop innovative solutions that go beyond traditional algorithmic methods. It involves using imagination to explore multiple perspectives, synthesise diverse ideas, and design original approaches to solving problems. Creativity empowers learners to develop unique solutions, optimise processes, and create applications that address real-world challenges in fields like software development, artificial intelligence, cybersecurity, and data science.	

Curriculum Resources

Lower Primary

Computational Thinking & Algorithms

- Bee-Bot & Blue-Bot – Physical coding toys for sequencing and problem-solving (reference the app too in this)
- Barefoot Computing – Free resources for computational thinking activities (barefootcomputing.org)
- https://code.org/student/elementary *(only GDPR compliant without a login)*
- https://codejr.org/scratchjr/
- https://compute-it.toxicode.fr/ - online platform that teaches computational thinking and programming concepts through puzzle-based challenges.

Programming & Coding Basics

- Ozobot (hardware in combination with https://ozoblockly.com/editor) - colour-based programming and block-based coding *(only GDPR compliant without a login)*
- LEGO WeDO – Simple robotics and coding for young learners
- Lego Spike Essential

Digital Literacy

- TypingStudy – Support for learning how to type on a computer (http://typingstudy.com/)
- Typing.com *(only GDPR compliant without a login)*
- Typing Club *(only GDPR compliant without a login)*

Online Safety

- Common Sense Education – Digital citizenship and online safety (commonsense.org/education)
- Kapow – Support resources and training on how to teach and deliver online safety in primary school settings
- https://saferinternet.org.uk/

Digital Creativity & Presentations

- Book Creator – Basic digital storytelling and presentation creation
- Sketches for School – Creative tools for drawing and basic digital art
- Chrome Music Lab
- https://www.w3schools.com/
- Visual Studio Code
- https://saferinternet.org.uk/

Web Development:

- https://code.org/educate/weblab *(only GDPR compliant without a login)*
- https://www.w3schools.com/
- Visual Studio Code
- Chrome Music Lab

Upper Primary

Computational Thinking & Algorithms

- Barefoot Computing – Advanced activities in computational thinking (barefootcomputing.org)
- https://vr.vex.com/ - online platform for programming virtual robots, teaching coding and computational thinking through interactive simulations.
- https://makecode.microbit.org/

Programming & Coding

- Kai´s Clan – STEM toolset which allows for multiplayer coding (https://robotixeducation.ca/collections/kais-clan)
- Sphero – STEM learning company with specialist resources on Robotics (https://sphero.com/)
- Swift Playgrounds – Interactive iPad coding using Swift
- LEGO SPIKE Prime – Advanced robotics and coding integration
- Micro :bit – Physical computing and basic electronics projects
- Code.org – Structured courses introducing programming and problem-solving (code.org) *(only GDPR compliant without a login)*
- Sonic Pi

Data Representation & Digital Literacy

- BBC Bitesize KS2 Computer Science – Basic theory and quizzes (bbc.co.uk/bitesize)
- Microsoft Excel / Numbers – Basic data handling, charts, and introductory formulas

Digital Creativity & Presentations

- Book Creator – Digital storytelling and presentation creation
- Tinkercad – Intro to 3D design and modelling
- Canva – Graphic design and presentation tool
- PowerPoint – Slide-based presentation tool
- Photoshop – Photo editing and design tool
- Adobe Creative Cloud – Suite of creative design tools
- iMovie – Free Video Editing Application
- https://vr.vex.com/ - online platform for programming virtual robots, teaching coding and computational thinking through interactive simulations.
- https://hackidemia.github.io/cognimates-gui/ - is an AI-powered platform designed for educational purposes, enabling users to create projects, train AI models, and explore machine learning through a block-based interface.
- https://makecode.microbit.org/
- Tinkercad
- Canva – Graphic design and presentation tool
- PowerPoint – Slide-based presentation tool
- Photoshop – Photo editing and design tool
- Adobe Creative Cloud – Suite of creative design tools
- iMovie – Free Video Editing Application
- Lunapic photo editor

Lower Secondary

Computational Thinking & Algorithms

- Teach-ICT.com – Detailed theory resources for KS3
- BBC Bitesize KS3 Computer Science – Comprehensive theory and quizzes (bbc.co.uk/bitesize)

Programming & Coding

- Python.org (Docs & Tutorials) – Official Python documentation and beginner tutorials (python.org)
- W3Schools – Interactive coding tutorials (Python, HTML, CSS, JavaScript) (w3schools.com)
- Replit – Cloud-based IDE for Python, HTML/CSS, and JavaScript (replit.com)
- Visual Studio Code – Advanced text editor for console-based and web-stack development

Data Representation, Systems & Networks

- Logic.ly – Logic gate simulations (Free trial available)
- Teach-ICT.com – Resources on data representation, computer systems, and networks (staff only)

Robotics & Hardware Integration

- LEGO Mindstorms – Advanced robotics and programming integration
- Micro:bit – Physical computing and complex electronics projects

Digital Literacy & Productivity

- Microsoft Word / Pages – Advanced word processing and document creation
- Microsoft Excel / Numbers – Advanced data handling, formulas, and data analysis
- Keynote / PowerPoint – Professional presentations with multimedia integration
- Photoshop – Advanced image editing and digital art creation

Digital Creativity & Web Development

- TinkerCAD – Advanced 3D design and modelling
- Visual Studio Code & Replit – Full-stack web development (HTML, CSS, JavaScript)

Prohibited System Usage (GDPR Concerns)

Various systems above are indicated as only being compliant when used without a login. Additionally, Web Lab and Scratch (full version) are not compliant if used with a login.

Lower Primary Knowledge and Skills Objectives

Programming & Computational Thinking

Objective	Concept	Skill
Understand what algorithms are; how they are implemented as programs on digital devices; and that programs execute by following precise and unambiguous instructions	Logic, Form, Function	Programming, Computational Thinking
Create and debug simple programs	Procedure, Adaptation	Programming, Computational Thinking
Use logical reasoning to predict the behaviour of simple programs	Logic, Function	Programming, Computational Thinking Creativity

Digital Literacy

Objective	Concept	Skill
Understand how to use age-appropriate software for word processing, data handling, and presentations.	Form, Function	Digital Literacy, Creativity, Data Handling
Use digital tools to create and edit simple text, images, and shapes in documents and presentations.	Procedure, Adaptation	Digital Literacy, Creativity
Save, open, and organise digital work with support, understanding the importance of keeping files safe.	Procedure	Digital Literacy
Select and use suitable digital tools for writing, creating pictures, or presenting information.	Procedure, Innovation	Digital Literacy, Creativity
Use technology purposefully to create, organise, store, manipulate, and retrieve digital content.	Data, Innovation	Digital Literacy
Recognise common uses of information technology beyond school.	Perspective, Data	Data Handling, Digital Literacy
Use technology safely and respectfully, keeping personal information private; identify where to go for help and support when they have concerns about content or contact on the internet or other online technologies.	Perspective, Data	Digital Literacy, Data Handling
Familiarise with input devices such as keyboard, mouse, and touchscreen (iPad), and understand their use.	Function, Procedure	Digital Literacy
Make decisions on the most suitable digital tools to complete a project.	Perspective, Procedure	Digital Literacy
Begin using simple software to create and edit images and shapes (e.g. Sketches for School).	Procedure, Adaptation	Digital Literacy, Creativity
Use age-appropriate search technologies and research skills.	Perspective, Data	Digital Literacy

Curriculum adaptations

Objective	SGS Objective	Rationale
N/A	Use technology safely and respectfully, keeping personal information private; identify where to go for help and support when they have concerns about content or contact on the internet or other online technologies.	Additional LO in order to support student understanding of data in line with future requirements within a German setting.
N/A	Understand how to use age-appropriate software for word processing, data handling, and presentations.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Use digital tools to create and edit simple text, images, and shapes in documents and presentations.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Save, open, and organise digital work with support, understanding the importance of keeping files safe.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Select and use suitable digital tools for writing, creating pictures, or presenting information.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Use technology purposefully to create, organise, store, manipulate, and retrieve digital content.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Recognise common uses of information technology beyond school.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Use technology safely and respectfully; keep personal information private; identify where to go for help when there are concerns about content.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Familiarise with input devices such as keyboard, mouse, and touchscreen (iPad), and understand their use.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Make decisions on the most suitable digital tools to complete a project.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Begin using simple software to create and edit images and shapes (e.g., MS Paint/Sketches for School).	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.
N/A	Use age-appropriate search technologies and research skills.	Additional LO in order to build Digital Literacy skills for lower primary students, building into skills for Upper Primary.

Upper Primary Knowledge and Skills Objectives

Programming & Computational Thinking

Objective	Concept	Skill
Design, write and debug programs that accomplish specific goals, including controlling or simulating physical systems; solve problems by decomposing them into smaller parts	Procedure, Adaptation	Programming, Computation Thinking
Use sequence, selection, and repetition in programs; work with variables and various forms of input and output	Form, Function	Programming, Computational Thinking, Digital Literacy
Use logical reasoning to explain how some simple algorithms work and to detect and correct errors in algorithms and programs	Logic, Adaptation	Programming, Computational Thinking
Select, use and combine a variety of software (including internet services) on a range of digital devices to design and create a range of programs, systems and content that accomplish given goals, including collecting, analysing, evaluating and presenting data and information	Form, Data	Digital Literacy, Creativity

Digital Literacy & Online Safety

Objective	Concept	Skill
Independently combine tools to create multimedia presentations, incorporating animations, images, and sound.	Procedure, Adaptation	Creativity Digital Literacy
Manage files independently across different cloud platforms and devices, ensuring responsible data organisation.	Procedure	Digital Literacy, Data Handling
Explore how different careers and industries use technology, including coding, AI, and automation.	Perspective	Digital Literacy, Computational Thinking
Use advanced search techniques to evaluate credibility, sources, and understand online bias.	Perspective, Data	Digital Literacy, Research Skills
Analyse online risks, including cyberbullying, misinformation, and privacy concerns, and develop strategies to manage them.	Perspective, Data	Digital Literacy
Design and present multimedia projects with a focus on audience engagement and ethical digital content use.	Digital Literacy	Digital Literacy, Creativity
Improve speed and accuracy in typing, navigating multiple interfaces, and using shortcut functions.	Procedure	Digital Literacy

Curriculum Adaptations

Objective	SGS Objective	Rationale
Use search technologies effectively, appreciate how results are selected and ranked, and be discerning in evaluating digital content	Use simple search technologies to find age-appropriate information and images online, with guidance. Begin to understand that not all online information is reliable and that personal data should not be shared.	Introduces awareness of data privacy and the influence of practices relevant to the German context.
N/A	Independently combine tools to create multimedia presentations, incorporating	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare

	animations, images, and sound.	students for more advanced applications in Lower Secondary
N/A	Manage files independently across different cloud platforms and devices, ensuring responsible data organisation.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary
N/A	Explore how different careers and industries use technology, including coding, AI, and automation.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary
N/A	Use advanced search techniques to evaluate credibility, sources, and understand online bias.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary
N/A	Analyse online risks, including cyberbullying, misinformation, and privacy concerns, and develop strategies to manage them.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary
N/A	Design and present multimedia projects with a focus on audience engagement and ethical digital content use.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary
N/A	Improve speed and accuracy in typing, navigating multiple interfaces, and using shortcut functions.	Develop progressive Digital Literacy skills that build upon foundational knowledge in Lower Primary and prepare students for more advanced applications in Lower Secondary

Lower Secondary Knowledge and Skill Objectives

Computational Thinking

Objective	Concept	Skill
Model real-world problems and physical systems using computational abstractions, focusing on the impact of emerging technologies in a global context	Form, Innovation	Computational Thinking, Creativity in Technology
Understand several key algorithms that reflect computational thinking (e.g., sorting and searching); use logical reasoning to compare the utility of alternative algorithms for the same problem	Logic, Process	Computational Thinking, Programming
Apply abstraction and pattern recognition to create efficient and scalable solutions for complex computational problems.	Logic, Process	Computational Thinking
Analyse the efficiency of algorithms using Big-O notation (time and space complexity)	Logic	Computational Thinking
Investigate basic encryption methods (e.g., Caesar cipher) and explain their application in securing data.	Innovation	Programming

Programming & Development

Objective	Concept	Skill
Use multiple programming languages to solve computational problems; design and develop modular programs that use efficient data structures, such as lists, arrays, or linked lists, for scalable solutions	Procedure, Data, Adaptation	Programming, Data Handling
Understand simple Boolean logic (e.g., AND, OR, NOT) and some of its uses in circuits and programming	Logic	Programming
Design, implement, and test recursive algorithms for solving computational problems.	Procedure	Programming
Optimise program code to improve its efficiency using techniques such as loop unrolling or dynamic programming.	Adaptation, Logic	Programming

Computer Systems & Data Representation

Objective	Concept	Skill
Understand how numbers can be represented in binary, and be able to carry out simple operations on binary numbers (e.g., binary addition, and conversion between binary and decimal)	Data, Logic	Programming, Data Handling
Understand the hardware and software components that make up computer systems, focusing on system architecture, network topologies, and the impact of cloud computing and distributed systems.	Innovation, Form	Computational Thinking
Understand how instructions are stored and executed within a computer system; understand how data of various types (including text, sounds and pictures) can be represented and manipulated digitally, in the form of binary digits	Data, Process	Programming, Data Handling
Understand the structure of computer systems, including the role of CPU, RAM, ROM, and storage devices, and how these impact performance.	Process	Computational Thinking
Understand binary representation, including negative binary numbers (two's complement), hexadecimal, and their applications in computer systems and networking	Data, Logic	Programming, Data Handling

Understand and apply binary and hexadecimal representations in data storage and networking, including conversions and operations.	Data	Data Handling

Digital Literacy & Online Safety

Objective	Concept	Skill
Undertake creative projects that involve selecting, using, and combining multiple applications, preferably across a range of devices, to achieve challenging goals, including collecting and analysing data and meeting the needs of known users	Innovation, Adaptation	Data Handling, Creativity in Technology
Create, re-use, revise and re-purpose digital artefacts for a given audience, with attention to trustworthiness, design and usability	Function, Adaptation	Digital Literacy, Creativity in Technology
Investigate and evaluate the use of artificial intelligence in digital systems, with a focus on recognising bias, assessing the reliability of sources, and applying critical thinking to ensure safe, ethical, and informed use of AI technologies.	Adaptation	Digital Literacy

Contemporary Issues

Objective	Concept	Skill
Explore the ethical issues in computing, including privacy, data security, and the impact of emerging technologies.	Data	Digital Literacy
Develop strategies to handle harmful digital content, including misinformation, cyberbullying, and digital addiction.	Data	Digital Literacy
Understand the relationship between screen time, digital addiction, and mental health, and develop strategies to maintain a healthy balance between online and offline activities.	Function, Adaptation	Digital Literacy
Understand the importance of online security and privacy, particularly in social media, personal data, and internet governance; recognise harmful digital behaviours (e.g., cyberbullying, misinformation) and develop strategies for digital citizenship	Innovation, Adaptation	Digital Literacy
Understand how artificial intelligence (AI) and automation are used in everyday applications like recommendation systems, virtual assistants, and machine learning, and explore the ethical implications of these technologies on society, including bias and fairness in AI algorithms.	Innovation, Adaptation	Digital Literacy, Computational Thinking
Explore the concept of a digital footprint and how online actions, such as social media posts and online interactions, can have long-lasting effects on personal reputation and future opportunities.	Data	Digital Literacy
Understand the rise of misinformation and fake news in digital media and social networks and develop skills to critically evaluate online sources of information.	Data	Data Handling
Evaluate the role of emerging technologies (e.g., AI, quantum computing, blockchain) and their societal and ethical implications.	Innovation, Function, Perspective	Digital Literacy
Understand the fundamentals of blockchain technology, including its applications in cybersecurity, finance, and data integrity.	Function, Adaptation	Digital Literacy, Data Handling
Analyse how digital systems process, store, and transmit data securely, including encryption and decentralised networks.	Logic, Form	Data Handling

Investigate digital footprints, personal data security, and the role of GDPR and privacy laws in protecting user rights.	Perspective, Data	Digital Literacy,
Develop strategies for **i**dentifying and managing misinformation, bias, and deepfakes in digital content.	Perspective, Innovation	Digital Literacy
Explore the impact of automation, robotics, and AI-driven decision-making **on** careers and society.	Form, Adaptation	Digital Literacy

Curriculum Adaptations

Objective	SGS Objective	Rationale
Design, use and evaluate computational abstractions that model the state and behaviour of real-world problems and physical systems.	Model real-world problems and physical systems using computational abstractions, focusing on the impact of emerging technologies in a global context	Both the IGCSE and IBDP curricula emphasise the need to understand the implications of emerging technologies globally. By highlighting real-world problem-solving, students can link their computational skills to global issues (e.g., climate change or global communication systems).
Use two or more programming languages, at least one of which is textual, to solve a variety of computational problems; make appropriate use of data structures (e.g., lists, tables or arrays); design and develop modular programs that use procedures or functions	Use multiple programming languages (including object-oriented and functional programming languages) to solve computational problems; design and develop modular programs that use efficient data structures, such as lists, arrays, or linked lists, for scalable solutions	Both the IGCSE and IBDP syllabi focus on object-oriented and functional programming. The use of languages like Python & Java, SQL, ~~and JavaScript is central~~, and students are expected to understand scalability and efficiency in their code
Understand the hardware and software components that make up computer systems, and how they communicate with one another and with other systems	Understand the hardware and software components that make up computer systems, focusing on system architecture, network topologies, and the impact of cloud computing and distributed systems.	Both the IGCSE and IBDP curricula emphasise cloud computing and distributed systems, such as server-client relationships and cloud storage. Focusing on these topics will better prepare students for advanced content. Adaptation for Germany: Given Germany's strong cybersecurity focus, it is important to include data security and ethical considerations in system design, particularly how GDPR impacts digital data storage and communication.
Understand a range of ways to use technology safely, respectfully, responsibly and securely,	Understand the importance of online security and privacy, particularly in social media, personal data, and internet	The IGCSE and IBDP curricula require students to be more aware of online ethics, security (including how to avoid cyber threats), and

including protecting their online identity and privacy; recognise inappropriate content, contact and conduct and know how to report concerns.	governance; recognise harmful digital behaviours (e.g., cyberbullying, misinformation) and develop strategies for digital citizenship	digital citizenship. In Germany, students should be aware of specific laws and regulations around privacy and digital footprints
Understand how numbers can be represented in binary, and be able to carry out simple operations on binary numbers (e.g., binary addition, and conversion between binary and decimal).	Understand binary representation, including negative binary numbers (two's complement), hexadecimal, and their applications in computer systems and networking	The IGCSE and IBDP focus on more complex forms of data representation, such as two's complement for signed binary numbers and hexadecimal numbers, particularly in networking and hardware contexts.
N/A	Explore the ethical issues in computing, including privacy, data security, and the impact of emerging technologies.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Develop strategies to handle harmful digital content, including misinformation, cyberbullying, and digital addiction.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Understand the relationship between screen time, digital addiction, and mental health, and develop strategies to maintain a healthy balance between online and offline activities.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Understand the importance of online security and privacy, particularly in social media, personal data, and internet governance; recognise harmful digital behaviours (e.g., cyberbullying, misinformation)	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping

	and develop strategies for digital citizenship	students with critical awareness and adaptability in an evolving technological landscape.
N/A	Understand how artificial intelligence (AI) and automation are used in everyday applications like recommendation systems, virtual assistants, and machine learning, and explore the ethical implications of these technologies on society, including bias and fairness in AI algorithms.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Explore the concept of a digital footprint and how online actions, such as social media posts and online interactions, can have long-lasting effects on personal reputation and future opportunities.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Understand the rise of misinformation and fake news in digital media and social networks and develop skills to critically evaluate online sources of information.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Evaluate the role of emerging technologies (e.g., AI, quantum computing, blockchain) and their societal and ethical implications.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Understand the fundamentals of blockchain technology, including its applications in cybersecurity, finance, and data integrity.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness

		and adaptability in an evolving technological landscape.
N/A	Analyse how digital systems process, store, and transmit data securely, including encryption and decentralised networks**.**	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Investigate digital footprints, personal data security, and the role of GDPR and privacy laws in protecting user rights.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Develop strategies for **i**dentifying and managing misinformation, bias, and deepfakes in digital content.	Including contemporary topics such as blockchain, AI, cybersecurity, and emerging technologies ensures students engage with real-world digital challenges. These areas impact privacy, data security, and ethical decision-making, equipping students with critical awareness and adaptability in an evolving technological landscape.
N/A	Investigate and evaluate the use of artificial intelligence in digital systems, with a focus on recognising bias, assessing the reliability of sources, and applying critical thinking to ensure safe, ethical, and informed use of AI technologies.	Although not explicitly included in the UK national curriculum, this learning objective has been added to address the increasing relevance of artificial intelligence in students' lives. Developing critical awareness of AI and its biases, ethical implications, and safe use is essential for preparing learners to navigate and contribute responsibly to a rapidly evolving digital world.